IS1203: Foundations of Information Systems

Topic 3:
Part 2- Different Types of BIS

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Recommended Text:
https://paginas.fe.up.pt/~acbrito/laudon/index.htm
http://wps.prenhall.com/bp_laudon_essmis_6/21/5555/1422269.cw/-/t/index.html

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Learning Outcomes

- At the end of this lesson, you will be able to
  - Identify different types of systems that link the enterprise and improve organizational performance
  - Give examples to illustrate the functionalities of different types of enterprise applications
  - Describe the functionalities of customer relationship management, enterprise resource management and supply chain management systems.

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Enterprise Applications

- Span over different functional areas of an enterprise
- Focus on executing business processes across the business firm
- Include all levels of management
- Help businesses become more flexible, and productive by coordinating business processes more closely and integrating groups of processes for efficient management of resources and customer service

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4 Major Types of Enterprise Applications

- Enterprise Systems (Enterprise Resource Planning Systems)
- Supply Chain Management Systems
- Customer Relationship Management Systems
- Knowledge Management Systems

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Enterprise Systems

Also called Enterprise Resource Planning Systems - ERP Systems

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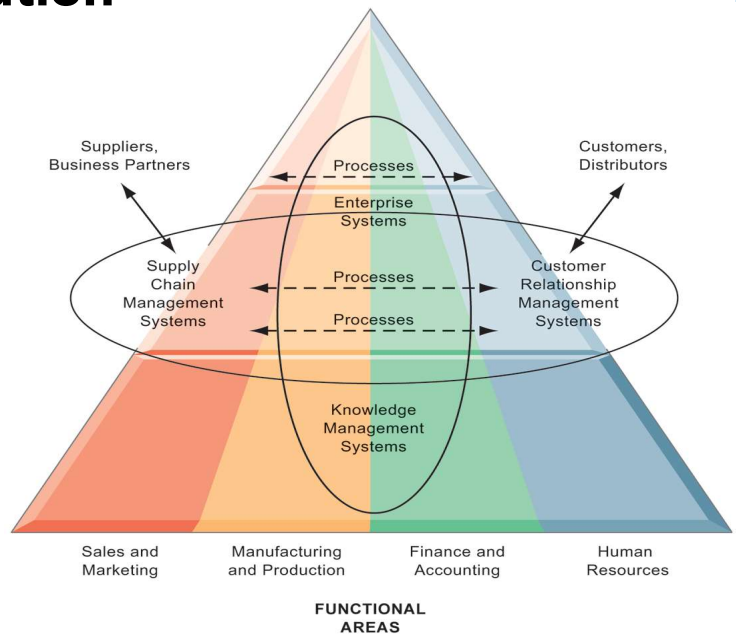
Enterprise Application Architecture

Enterprise applications automate processes that span multiple business functions and organizational levels and may extend outside the organization.



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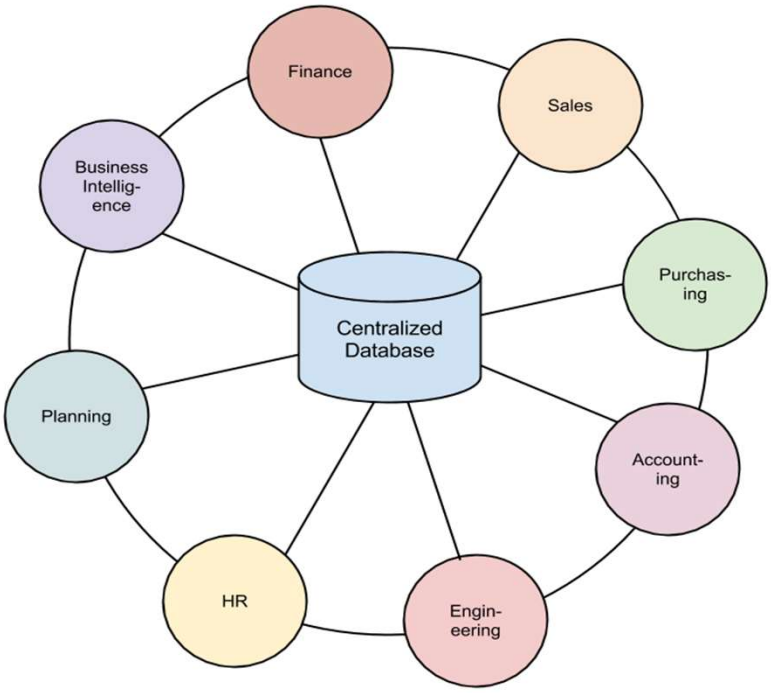
ERP System

A suite of integrated software modules and a common central database



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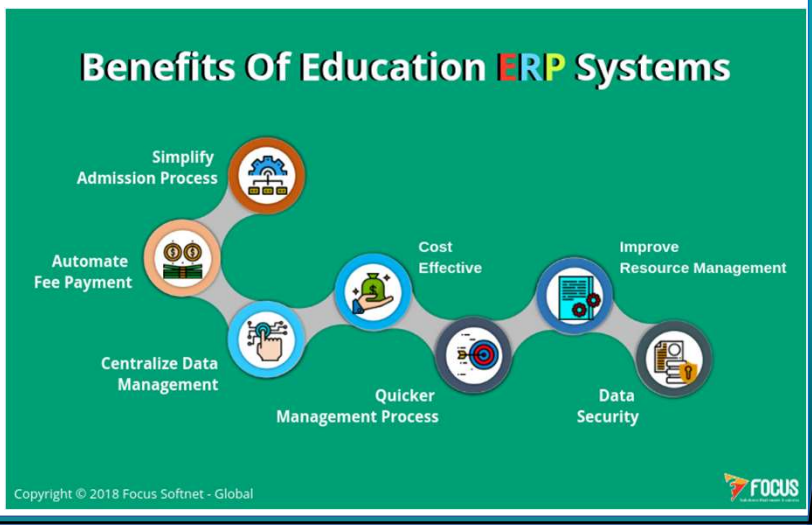
Benefits of ERP

- Increase productivity
- Decrease costs
- Enhance reporting
- Increase the quality and efficiency
- Support decision making
- Improve business processes



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Examples

- Acumatica Cloud ERP
  Deltek ERP
  Microsoft Dynamics 365 ERP
  ECount ERP
  Infor ERP
  JDEdwards EnterpriseOne
  Netsuite ERP
  Odoo ERP
- Paragon ERP
  Peoplesoft ERP
  Priority Software ERP
  SAP S4/HANA
  Sage ERP
  Syspro ERP

Sage 100cloud | EFI Radius | Oracle ERP | Abbasoft NetPartner

Choosing a suitable ERP system for your company
https://www.g2.com/categories/erp-systems/small-business

From https://www.itbusinessedge.com/articles/best-erp-systems-flexible-customizable.html

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Example ERP Systems… contd.

Company	Description	Features
Netsuite ERP	Market-leading cloud EPR solution. Modern and scalable implementation across all critical back-office functions.	Accounting, Inventory Management, Reporting & Analytics, & Functionality across enterprises of all sizes.
SAP ERP	Trusted EPR provider with 40 years of market history and more than 50,000 customers worldwide.	Accounting, Inventory & Warehouse Management, Purchasing, Reporting & Analytics Functionality across enterprises of all sizes.
Sage ERP	A fully-integrated, comprehensive suite of EPR business tools and capacity across all back-office functions.	Financial Cloud, Accounting, Inventory & Warehouse Management, Procurement, Reporting & Analytics, Functionality across enterprises of all functions.

From https://www.webopedia.com/TERM/E/ERP.html

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Exercise 1

Use https://www.g2.com and compare ERP solutions for the following.

1. Educational Institutions
2. Manufacturers
3. Hospital Management
4. Hotel Management

Use ChatGPT and identify the benefits an ERP can provide to each of the above contexts.

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Supply Chain Management (SCM) Systems

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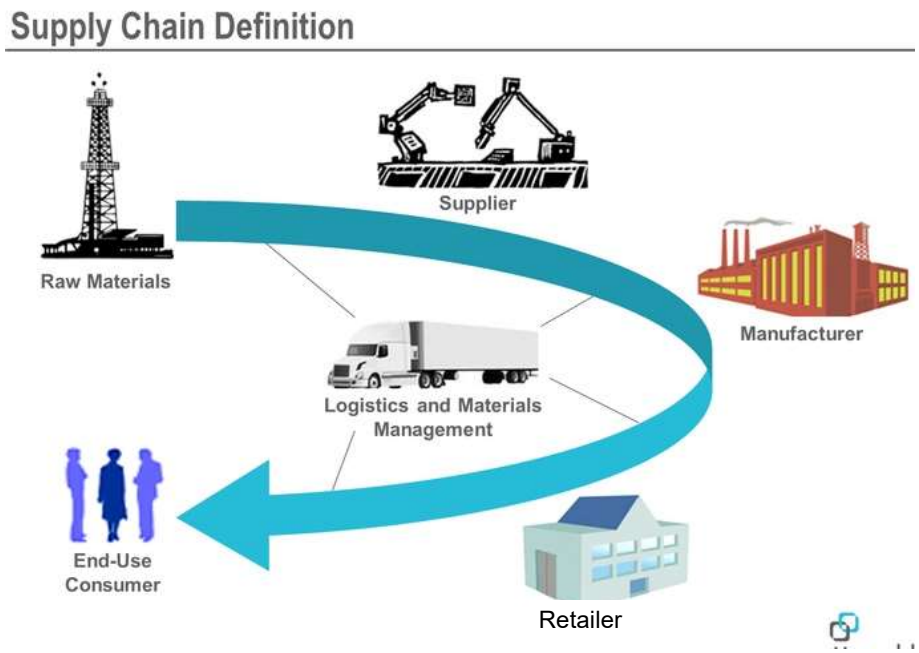

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An SCM System's Major Functions

1. Purchasing: The process of buying materials from suppliers for manufacturing products.
2. Operations: Activities a company does daily. E.g., Before stating production the company forecasts demand for its products
3. Logistics: Coordination of activities such as warehousing, inventory management, and transportation.
4. Resource management: Planning, organizing, and controlling of resources such as labour and raw materials.
5. Information workflow: A management function that relates to what and how information moves between members of the supply chain.

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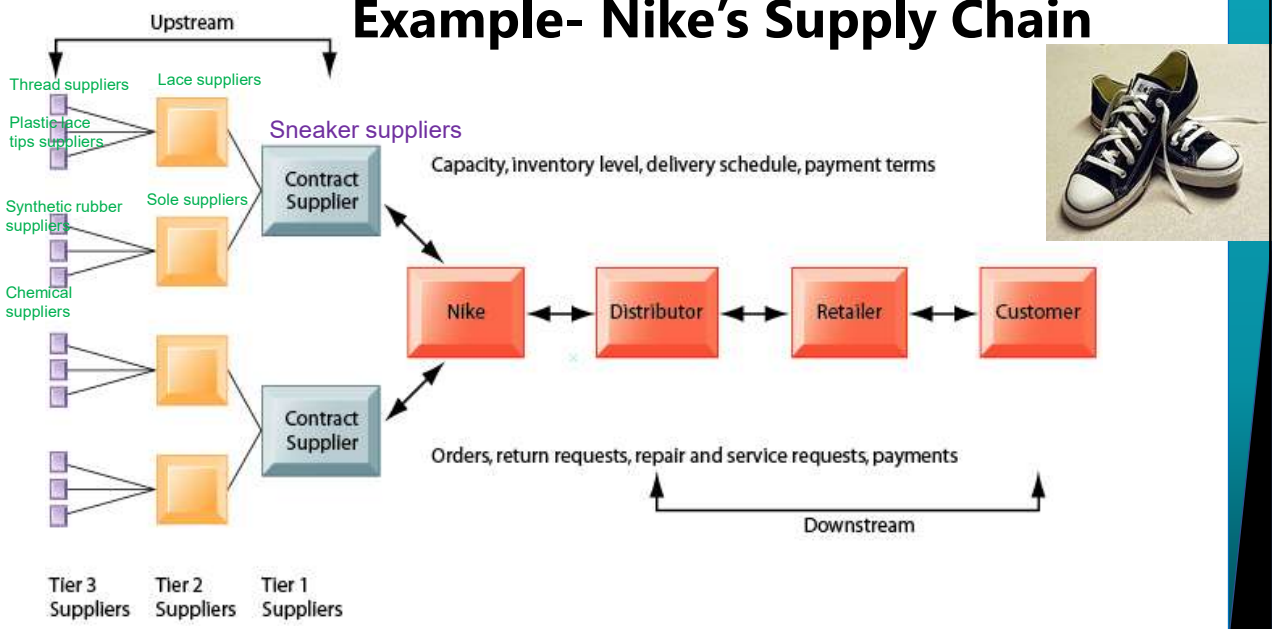
Example- Nike's Supply Chain



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Key Technologies in SCM Systems

1. **Enterprise Resource Planning (ERP) Integration** to ensure seamless data flow across supply chain and enterprise functions.
2. **Internet of Things (IoT): Use sensors and trackers** to monitor goods and equipment in real time and support decision-making.
3. **Advanced Analytics and Artificial Intelligence (AI):** Use AI and machine learning to forecast demand and optimize inventory, routing, scheduling, and resource allocation, and enhance decision-making.
4. **Blockchain Technology** to ensure secure and transparent transactions and traceability across the supply chain.
5. **Cloud Computing: Cloud-based** SCM solutions for greater scalability, flexibility, and access from anywhere.

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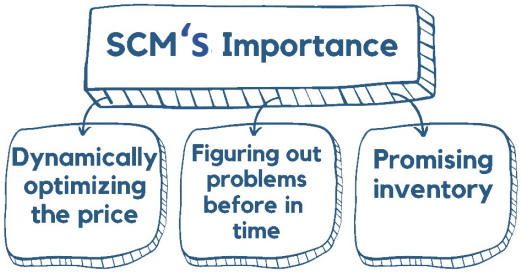
Business Value of SCM Systems

- Provide more accurate information on what to produce, store and move
- Can match supply to demand
- Reduce inventory level
- Improve delivery service
- Speed product time to market
- Use assets more effectively
- By reducing costs, it helps increase sales
- Make the right product available for customers to purchase at the right time



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Exercise 2

1. Define what enterprise applications are. Give three examples.
2. How do ERP systems help businesses achieve operational excellence?
3. Find possible reasons for ERP implementation failures.
4. Discuss the challenges of implementing SCM systems.
5. List major functionalities of an SCM system.

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Customer Relationship Management (CRM) System

A cross-functional system to integrate and automate many of the customer-serving processes.

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Customer Relationship Management

- Customer centric
  - Customer relationship is the most valued asset
  - Every company tries to retain the most valued customers
- CRM involves
  - Providing customer-facing employees with a single, complete view of every **customer at every touch point and across all channels**
  - Providing the customer with a single, complete view of the **company and its extended channels**

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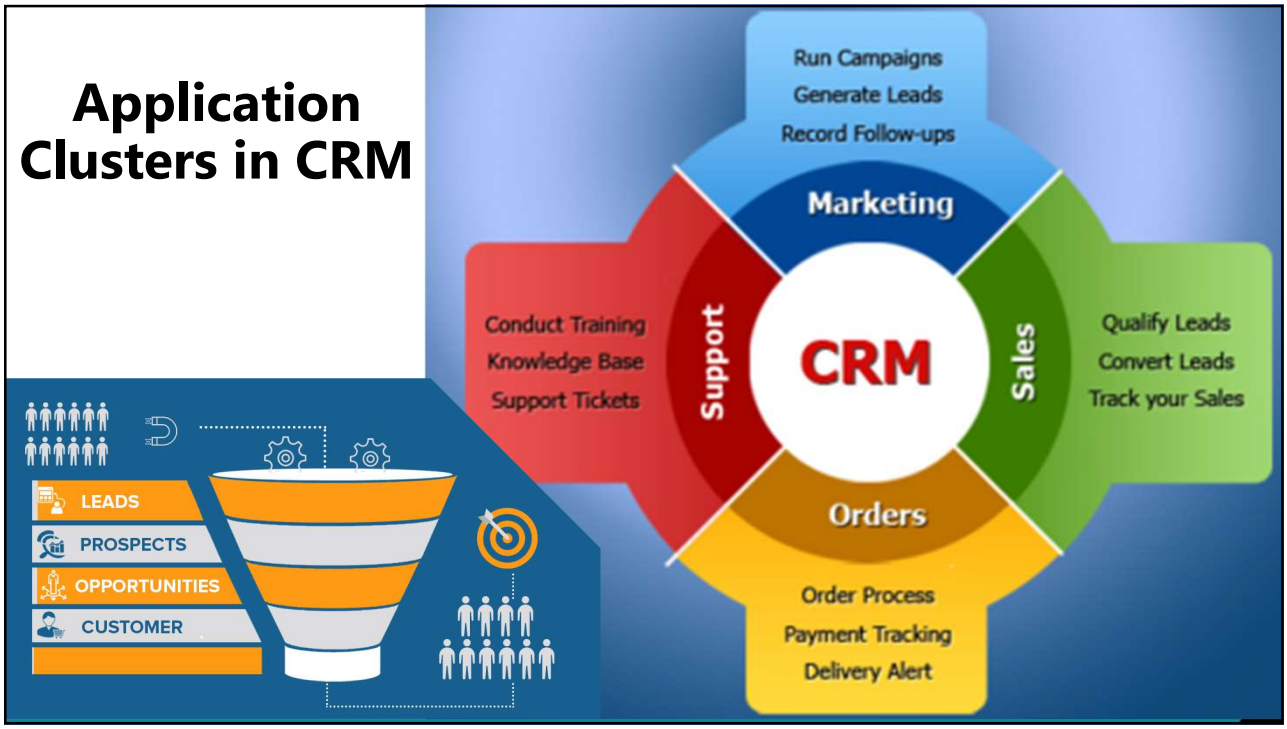
Application Clusters in CRM



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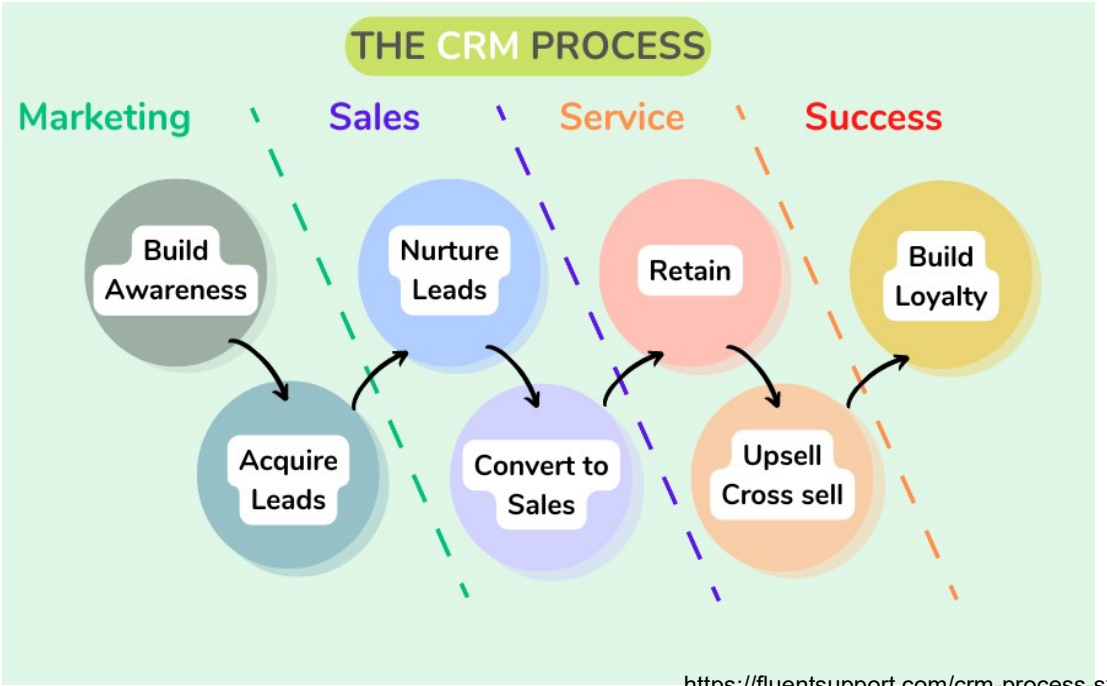

https://fluentsupport.com/crm-process-steps/

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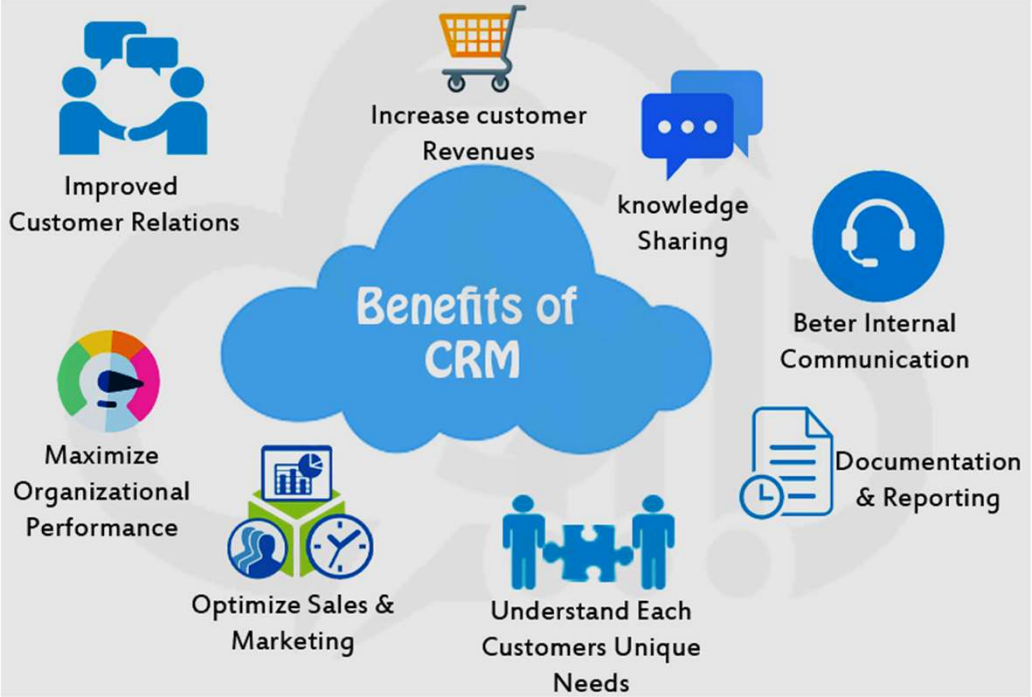

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e.g., Microsoft Dynamics 365

- https://www.youtube.com/watch?v=zyrBK6gJOHU

Microsoft Dynamics 365 is a cloud-based business applications platform that **combines components of customer relationship management (CRM) and enterprise resource planning (ERP), along with productivity applications and artificial intelligence tools**.

https://www.techtarget.com/searchcustomerexperience/definition/Microsoft-Dynamics-365

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Knowledge Management Systems (KMS)

- Collect internal knowledge and experience within the firm.
- Link to external sources of knowledge.
- Support processes for acquiring, creating, storing, distributing, applying, and integrating knowledge.
- Make the knowledge available where and whenever it is needed to perform duties, improve business processes and take management decisions.

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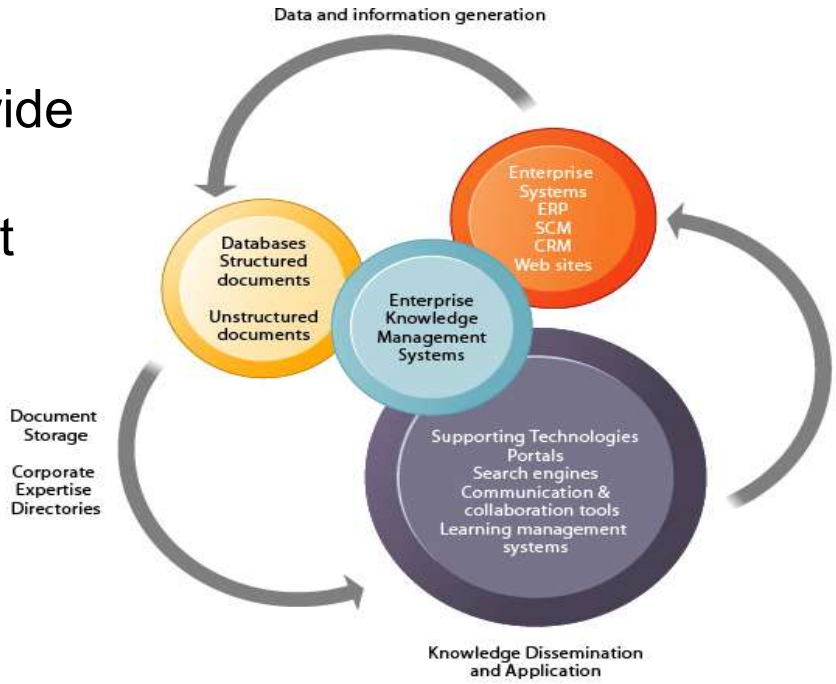
Enterprise-wide Knowledge Management



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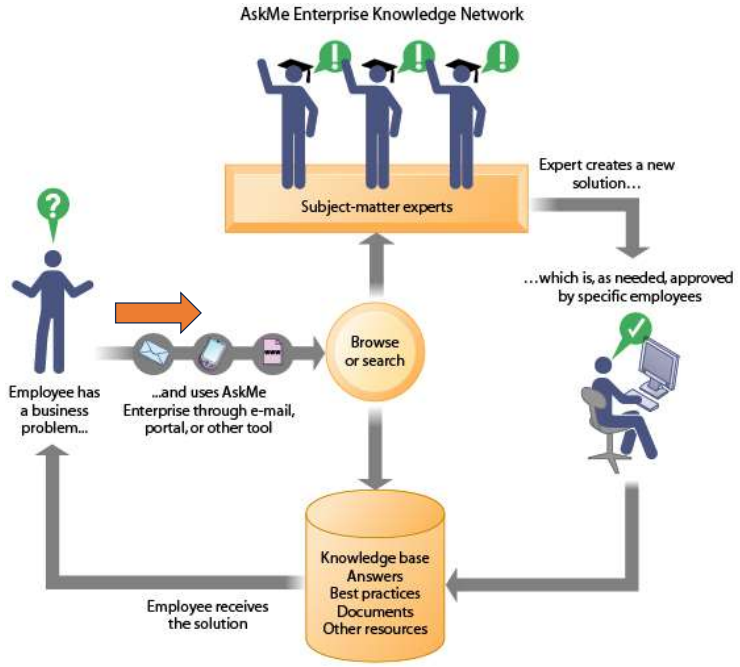
Knowledge Network System



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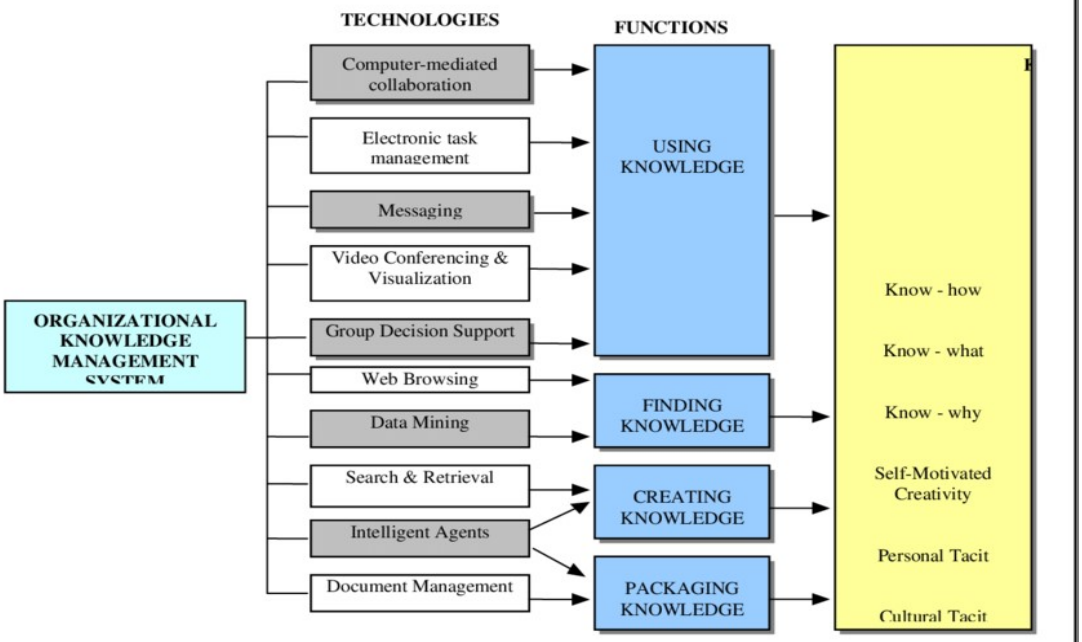

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Exercise 3

- Describe how knowledge can be stored in an organization.
- Explore the functionalities of the KMS using www.g2.com (https://www.g2.com/categories/knowledge-management)

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